



Review

The role of food chain in antimicrobial resistance spread and One Health approach to reduce risks

Prarthi Sagar^a, Ajmal Aseem^a, Santosh Kumar Banjara^{b,*}, Shobi Veleri^{a,**}^a Drug Safety Division, ICMR-National Institute of Nutrition, Hyderabad 500007, India^b Clinical Division, ICMR-National Institute of Nutrition, Hyderabad 500007, India

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ABSTRACT

The incidence of antimicrobial resistance (AMR) is rapidly spreading worldwide. It is depleting the repertoire of antibiotics in use but the pace of development of new antibiotics is stagnant for decades. Annually, millions of people are killed by AMR. This alarming situation urged both scientific and civil bodies to take steps to curb AMR as a top priority. Here we review the various sources of AMR in the environment, especially focusing on the food chain. Food chain inculcates pathogens with AMR genes and serves as a conduit for its transmission. In certain countries, the antibiotics are more used in livestock than in humans. It is also used in agriculture crops of high value products. The indiscriminate use of antibiotics in livestock and agriculture increased rapid emergence of AMR pathogens. In addition, in many countries nosocomial settings are spewing AMR pathogens, which is a serious health hazard. Both the developed and low and middle income countries (LMIC) face the phenomenon of AMR. Therefore, a comprehensive approach for monitoring all sectors of life is required to identify the emerging trend of AMR in environment. AMR genes' mode of action must be understood to develop strategies to reduce risk. The new generation sequencing technologies, metagenomics and bioinformatics capabilities can be resorted to quickly identify and characterize AMR genes. The sampling for AMR monitoring can be done from multiples nodes of the food chain as envisioned and promoted by the WHO, FAO, OIE and UNEP under the One Health approach to overcome threat of AMR pathogens.

1. Introduction

We live in an environment densely populated with microorganisms. They are present in almost all niches on the planet, including in the extreme conditions like acidic springs (Baker-Austin and Dopson, 2007) and on radioactive waste (Jeong and Choi, 2020). The bacteria are the second largest biomass on earth (2×10^{30}) (Flemming and Wuertz, 2019) after plants (Bar-On et al., 2018). They are necessary for many fundamental steps in ecosystem, e.g. to recycle the waste produced by the living organisms and for degrading organic debris to inorganic matter. Humans also require some of them as commensals in the body, e.g. *Bifidobacterium* spp. in gastrointestinal tract (GI) generates metabolites like short chain fatty acids from polysaccharides (Morrison and Preston, 2016). On the other hand, some of them are pathogenic and cause infectious diseases in organisms. In nature, some organisms like fungi limit the growth of bacterial colony in their niche by secreting

certain chemicals that inhibit metabolic process of bacteria. For example, *Penicillium notatum* produces penicillin, the first antibiotic discovered (Fleming, 2001). Dorothy Hodgkin in 1945 resolved that it has a β -lactam structure (Hodgkin, 1949). It triggered research into development of many semi-synthetic antibiotics like ampicillin. We have a large repertoire of antibiotics to treat variety of microbial pathogens. Most of these are natural product based antibiotics, which are mainly produced by actinomycetes - a filamentous bacteria- (64 %), followed by other bacteria and fungi (Hutchings et al., 2019).

In the past decades, indiscriminate and irrational continuous use of antibiotics in clinics and in livestock for controlling infections led the pathogens to evolve resistance to one or more antibiotics, e.g. in the case of multi-drug resistance tuberculosis (MDR-TB) (Khawbung et al., 2020). The antimicrobial resistance (AMR) becomes a global concern (Thompson, 2022). Increase in AMR at an alarming rate has been predicted to create serious health risks worldwide. An estimate in 2019

* Corresponding author.

** Correspondence to: S. Veleri, Drug Safety Division, ICMR-National Institute of Nutrition, Department of Health Research, Ministry of Health & Family Welfare, Govt. of India, Hyderabad PIN 500 007, India.

E-mail addresses: santosh.kumar@icmr.gov.in (S.K. Banjara), shobi.veleri@icmr.gov.in (S. Veleri).<https://doi.org/10.1016/j.ijfoodmicro.2023.110148>

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recorded 1.27 million deaths worldwide related to AMR compared with 8.64 million for HIV/AIDs and 6.43 million for malaria (Mensah et al., 2015). An estimate suggested that 10 million people could die yearly due to AMR by 2050 and it could overtake the number of deaths due to cancer now (O'Neill, 2016). The necessity of antimicrobial agents is widely recognized because of their ability to cure infectious diseases in humans and animals. Asia hosts 56 % of pigs and 54 % of chickens of the world and not surprisingly Asia is at the forefront for AMR in animals (Van Boeckel et al., 2019). In China and India the antibiotics use is increasing in livestock than in human (Van Boeckel et al., 2015). In global scale, 73 % of antibiotics sold are used in livestock reared for food (Van Boeckel et al., 2015; Van Boeckel et al., 2019). Livestock is a good source for protein rich food and in recent decades the demand for protein food has increased. This has concomitantly increased the use of antibiotic use in livestock to control infection and increase the yield of production. Thus, both clinical and livestock fronts are potential arena for evolving antibiotic resistance genes. They serve as entry points for AMR to reach food chain and re-circulate in the ecosystem. The three major factors contributing to AMR crisis are: 1). Microbes constantly exposed to antimicrobials evolve changes in genetic constitution and develop means to avoid the deleterious effect of antimicrobials. Such a capacity to overcome the effect of antibiotics on microbes is termed as antimicrobial resistance. It is a natural genetic trait inherent in organisms, and especially in the microbes that reproduce multiple generations within hours. 2). The technological advancement aiding rapid traveling, global trade, cold chain facilities and shipment enhanced worldwide spread of microbes, which might be carriers of AMR genes. 3). In the recent years, use of antimicrobials in poultry and livestock farming increased many folds (Michael et al., 2014). This sector is significantly contributing to the AMR besides the threat is emerging from the human health care sector in global scale.

Bacteria carrying AMR genes infecting animals and humans horizontally transmit it to the latter. However, the foremost route of transmission is through food that reaches GI tract. Bacteria with AMR could directly infect an organism or AMR mutations arising within the gut microbiota could spread via horizontal gene transfer (Collignon et al., 2016). The frequent use of antibiotics in hospitals to manage infections in patients provide ideal milieu for microbes to develop AMR, and for their rapid contamination to humans and food chain (D'Accolti et al., 2019). In Asia, Africa and South America 5–7 million people are annually infected with cholera and 100,000 are killed because of acute rehydration (Sack et al., 2004). This epidemic is treated with antibiotics like tetracycline, fluoroquinolones and azithromycin. The indiscriminate use of these antibiotics over the years led to development of resistance in cholera. The USA has witnessed a steady increase in isolates of resistant cholera from 3 % to 93 % from 1992 to 1994. It is a testimony for alarming rise of AMR across globe (Mahon et al., 1996). *Streptococcus pneumoniae* causes acute respiratory infections in children and >1 million children die every year due to these infections (Yamaguchi et al., 2021). This may be due to AMR in *S. pneumoniae* (Hedman et al., 2020). In EU infections due to AMR were highest in children below 1 year and adults above 65 years and high prevalence was in Italy and Greece (Cassini et al., 2018). Gram negative *Enterobacteriaceae* is resident of gut in animals and humans. Therefore, AMR emerging in this family of bacteria can easily spread across the environment through animal and human waste.

2. AMR in food chain

Food chain connects all the organisms in environment. The AMR elements in food horizontally transmit among the consumers of food pyramid. The vegetables, fruits, dairy products, milk, meat and other processed food products are rich source of AMR bacteria because antibiotics are widely used for prevention of infection and/or preservation of processed food (Acar and Moulin, 2006). Eating such food items horizontally transmits AMR to the consumers. Humans have strong link

in the food pyramid. Thus through food pyramid or by direct infection AMR genes enter human body. In addition, AMR genes can be contracted by contact with blood, feces, saliva harboring the pathogens. The contaminated foods, like dairy products, milk, meat, egg, seafood consumption is a major route for acquiring AMR genes (Jans et al., 2018). The lack of proper surveillance for AMR in food chain, especially in developing countries, is a security challenge for farmers and food handlers. They are at risk because of their continuous contact, directly or indirectly, with potential pathogens as carriers of AMR genes. They should be under proper surveillance to isolate and secure the farm to fork food chain from the risk of AMR (EFSA European Food Safety, 2014). The infected animals and persons could also vertically transmit AMR genes to their offspring. If a mother acquired AMR during pregnancy it could be transmitted to the child.

In environment, huge amount of antibiotic residues, excreted by organisms, remain in active form. It can be acquired by other organism in the food chain and is the most prominent reason for spreading AMR among communities (Thanner et al., 2016). Farm soil, manure, and waste water are identified hotspots of antibiotic resistant species. The residual antibiotic could elicit allergic reactions, cytotoxicity, mutagenicity, carcinogenicity, and its sustained endurance by the bacteria may lead to antibiotic resistance (Mensah et al., 2015). In Low and Middle Income Countries (LMICs), unprocessed waste of animal and human is used as fertilizer, which increases chance for horizontal transfer of AMR gene. Environmental pollution also plays major role in horizontal gene transfer.

2.1. Prevalence of AMR in non-vegetarian food

Globally, 22 % (1.5 billion) of people are vegetarians. In India 20–39 % of people are vegetarians (Wunsch, 2022). Taken together this suggests that 78 % of the world population is non-vegetarians. To meet such a huge demand for meat production antibiotics are extensively used in livestock farming more than in vegetable crop farming. The microbes in niche enter the food, which directly find way into GI tract. Therefore, agricultural produces, and food of animal origin, and processed foods, meat products and seafood inherently have the chance of containing microbes. Some of them may be a carrier for AMR genes. Approximately 50 % of foods of meat and seafood origin have contamination of bacteria like *Campylobacter*, *Enterococcus*, *Escherichia coli*, *Listeria*, *Salmonella*, and *Vibrio* spp., which are increasingly resistant to antibiotics. Gram negative bacteria are resistant to antibiotics like aminoglycosides, cephalosporins, fluoroquinolones, penicillin, sulfonamides and tetracyclines (Jans et al., 2018). Gram positive bacteria like *Staphylococcus* and *Enterococcus* are resistant to glycoproteins, lincosamides, macrolides and nitrofurans, and are found in meat sources, while *Staphylococcus* is detected in seafood, species like *Enterococci* are safely used for starter culture in fermentation in dairy industry (Jans et al., 2018).

In developing countries, the clinical use of antibiotics is not stringently regulated either for humans or animals; even in food processing sectors antibiotics usage is weakly regulated. Generally, in LMICs proper monitoring of antibiotics administration and usage is lacking. Therefore, reliable data on AMR is limited in LMICs. In South Africa 1500 tonnes of antibiotics were used for livestock production in 2002 to 2004. Brazil, Chile, China and India reported 99 % increase in usage of antibiotics in proportion to population size (Van Boeckel et al., 2015). The developed countries are also no exception for AMR risk, e.g. transmission of AMR bacteria from broiler chickens to farmers has been detected (ECDC, 2016), which led to the 2006 ban on indiscriminate use of antibiotics in European Union though these studies were essentially focused on growth promoters. Total decrease in the usage of antibiotics from 1994 to 2014 has been 38 %. In 2014, Denmark and the Netherlands reported 2 % and 3 % reduction in use of antibiotics in livestock, respectively. The antibiotics use in livestock in EU has reduced 18.3 % in 2021 with reference to 2018 data (EMA/795956/2022, 2022). Penicillins were sold the maximum (31.2 % of prescribed antibiotics) in EU for livestock

use among the highest selling antibiotics.

Whereas the trend of antimicrobial use (AMU) in Denmark and the Netherlands, at community and hospital care settings reduced from 22.31 to 18.1 from 2012 to 2021, respectively; for Denmark it was 22.31 (ECDC, 2022). In 2021, the antibiotics use in hospital sector alone was 3.69 %, and the compound annual growth rate of use in EU increased 11.7 % (ECDC, 2022). The AMU in community level also increased marginally in the last ten years though there was no big change in last five years (ECDC, 2022).

3. AMR in agricultural crops

The crops constantly encounter various pests and pathogens in environment. This necessitates farmers to use pesticides and antibiotics for crop cultivation (Reeder, 2020). Antibiotics are not routinely used for most crops because it increases production cost. Antibiotics are used in high-value fruit crops and are mostly recommended for top three crops like rice, tomato, and citrus. In crops like paprika, potato, cabbage, eggplants, pumpkins, onions and maize etc. antibiotics are less frequently used. Mostly antibiotics are recommended for treating bacterial infections, followed by in combinations to treat insects or mites, and then for fighting fungal infections. In South East Asia (SEA), the tropical environment is a thriving ground for many insects and mites, which feeds on plants. Therefore, 90 % of AMU is for treating these insects and mites. In SEA, the trend seems to use a combination of insecticide, fungicide and antibiotics.

The six antibiotics (streptomycin, tetracycline, oxytetracycline, gentamicin, cefadroxil, amoxicillin) used in agriculture practices could develop antimicrobial resistance and pose risk for the resources of human medicine. Streptomycin-resistant gene *strAB* was identified in *Erwinia amylovora*, *Pseudomonas syringae*, and *Xanthomonas campestris* etc. (Sundin and Wang, 2018). This is believed to be acquired from epiphytic commensal organisms in plants and a horizontal gene transfer to other bacteria because identical AMR gene carrying plasmids were isolated from humans and plants (Palmer et al., 1997) (Yau et al., 2010). Currently antibiotic use in crop is monitored in only few countries (FAO and WHO, 2019). A meagre 3 % of the 158 countries have a system to monitor antibiotics use in plants, while 23 % and 26 % of them have monitoring for antibiotics use for animal and human health, respectively. Moreover, reliable data for AMU in crops is unavailable. The most available data on AMU in crops were mainly obtained from apple and pear farms in the USA, where farmers used antibiotics to fight fire blight caused by *Erwinia amylovora*. Streptomycin use in these crops did not pose any known environmental risk (McManus, 2014). A similar result was reported from Switzerland too; though the authors could not exclude long term impacts on environment and on all fauna and flora (Gusberti et al., 2015). The use in crops was estimated to be 0.25 to 0.5 % in the USA, which is very low compared to the use in livestock and aquaculture (McManus, 2014). This may not pose a risk of AMR. Nevertheless, the AMU practice in crops varies significantly across different countries. The reported AMU in crops is almost nil in Africa and is reaching up to 2.5 % in SEA. In China AMU in crops reached up to 4.5 % (Tao Zhang et al., 2017). In special cases, for example, in Costa Rica, tetracycline and gentamycin were used in concentration of 200 to 700 times than for clinical purpose in human, and is a seriously alarming situation (Sánchez, 2008). The indiscriminate and deliberate spraying of antibiotics on crop plants introduces AMR risk in a wide range of environmental settings. Its long term impact is completely unknown. This has to be addressed at the policy level. SEA and Western Pacific (WP) approved use of AMU in crops but not in Brazil and EU. The crops like tomato and citrus are often eaten raw. If they carry bacteria with AMR genes and without proper wash eaten raw could potentially transmit the AMR genes into animals and humans (Walia et al., 2012).

Streptomycin ranks first in crop use, followed by tetracycline and kasugamycin. The usage of antibiotics has regional bias. In SEA streptomycin and tetracycline are frequently used. In this region, an

estimated 75 million hectare rice field used 63 tonnes of streptomycin and 7 tonnes of tetracycline. This highlights the huge quantity of antibiotics introduced in to environment and how it can build up AMR risk in long term. In America and WP, gentamycin and oxytetracycline are the main choice. Interestingly, in WP countries only oxolinic acid, ningnanmycin and validamycin are used. Likewise, in America used cefadroxil and Eastern Mediterranean countries used amoxicillin (Reeder, 2020). Thus the AMU pattern of each country could help strategize and devise method for identification of AMR in local settings. The enhanced antibiotics use increases probability to evolve AMR against it, and the accumulation in environment. The pattern of antibiotics use and prevalence of AMR should have some level of logical correlation. This information must be kept in mind while monitoring food chain, especially using meta-genomics methods to analyse various environmental settings.

4. AMR in livestock

The animal sourced foods supplement major portion of the protein requirement for millions of people across the world (Adesogana et al., 2020). Animal food includes seafood, fish, poultry, cattle, pigs etc. The LMICs with low resource settings started more intensive animal farming practices to meet the high demand for protein food. This phenomenon is correlating with increased purchasing power in LMICs. This in turn increased AMU in animal husbandry practices for increasing the productivity. The increase in AMU is correlating with emerging AMR in livestock animals (Van Boeckel et al., 2019). This is a potential source for transfer of AMR to humans. The epidemiology of AMR in livestock is an emerging area of research and it has recently been gained tremendous importance (Hedman et al., 2020). The AMR in pigs and broiler was increasing in Europe (Hesp et al., 2019). A Europe wide AMR surveillance in livestock is underway to study the epidemiology (Mader et al., 2022).

4.1. Seafood and inland fish

Seafood and fish are excellent precursors of low-fat high-quality proteins, omega-3 fatty acids or n-3 polyunsaturated fatty acids (PUFAs) and other nutrients such as minerals, vitamins and trace elements (Sidhu, 2003). Seafood is rich in vitamins like vitamin-A (retinol), -D (calciferol), -E (tocopherol), and B vitamins (niacin, thiamin, riboflavin, B6 and B12). Oily fish like salmon and sardines are a good source of vitamins A, -D, and -E. Seafood and fish are highly nutritious and are rich source of calcium, iodine, iron, magnesium, phosphorus, potassium, and zinc. It's rich macro- and micro-nutrients make them an important source of nutrition available easily across the globe (Arino et al., 2013). Eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) are the most important omega-3 fatty acids in seafood (Mozaffarian and Rimm, 2006). Every fish contains omega-3 fatty acids but fatty fish like herring, mackerel, salmon, sardines, trout, tuna, and oysters have high quantity of it. (Torris et al., 2018). These fatty acids can be beneficial to drop blood pressure and thereby reducing the burden of coronary heart disease, and preventing certain cardiac arrhythmias and sudden death (Sidhu, 2003). It is reported that the consumption of fatty fish reduces the risk of dementia by 28 % (Fotuhi et al., 2009) and fish oil is used for prevention of diabetes (Soltan and MEM, 2012).

Asia produces two-thirds of global fish demand (Schar et al., 2021). Aquaculture contributes a significant portion of animal protein (~17 %) and 6.7 % of protein consumed worldwide in 2013. Approximately 3.1 billion people get their protein requirement (20 % of daily average consumption) from fish (Guillen et al., 2018). The total fish production was 93.4 million tonnes in 2014; 81.5 million tonnes of marine and 11.9 million tonnes of inland sources (Guillen et al., 2018). From 2010 to 2018, the total fish production increased 1.2 million tonnes (148 to 178.8). China is the major fish producer that contributed 35 % of global fish production. India is at second position with 6 % of total fish

production of the world (9, 045,892 metric tonnes annually). Thus fish and seafood form a major portion of food consumed worldwide. In this regard, aquaculture plays a significant role to meet world food security demand and alleviate poverty. But the fish farms frequently report Edwardsiellosis and Streptococcosis. The acute hepatopancreatic necrosis is common in farm fish when temperature hikes. If unattended and untreated these diseases may affect the yield and income of the farmer. Therefore, farmers are often compelled to use antibiotics like tetracyclines, florfenicol, most β -lactams, quinolones, antifolates, chloramphenicol, erythromycin, and drug combination of trimethoprim and sulfametaxazol. *Edwardsiella* species are inherently resistant to fusidic acid, glycopeptides, lincosamides, macrolides, oxacillin, rifampin and streptogramins, and are widely used with minimum moderation (Reverter et al., 2020; Stock and Wiedemann, 2001).

Antimicrobials are mixed with fish feeds as means of in-feed medication, for example in cage farming fish (in open water resources of large reservoirs, sea and lakes). The unconsumed food and fish feces containing antibiotics settle at the bottom. Then antibiotics slowly leach and diffuse into the sediment and further drift away by water currents to other sites. It may even reach to marine water sources and ultimately accumulate in predatory wild fish and other organisms including shellfish. Thus antibiotics use could lead to wide spread contamination of the whole water body (Cabello, 2006).

The antibiotics presence in the aquatic environment can lead to AMR in pathogens of human and ultimately reach human microbiota (Cabello, 2006) (Fig. 1). For example, during the 1992 cholera epidemic in Latin America, AMR was spread via *Vibrio cholerae* through Ecuadorian shrimp industry because it was highly consumed by the local communities. Corrective policies should be implemented and strictly followed to decrease the level of AMR and spread to other environment as evidence indicates that AMR microbes and genetic elements could quickly transmit from the aquatic to the terrestrial environment (Cabello, 2006). Therefore, reducing the use of antibiotics and replacing them with improved sanitation practices in aquaculture to avoid the emergence of AMR in fish pathogens are necessary. For fish, pathogens causing infections in freshwater fish like *Aeromonas* species (*A. hydrophila*, *A. caviae*), *Pseudomonas* species (*P. fluorescens*, *P. putida*) *Flavobacterium* species (*F. branchiophilum*, *F. psychrophilum*), etc. antibiotics like oxytetracycline, sulfamerazin, flumequine, ciprofloxacin, sarafloxacin, erythromycin, clindamycin and vancomycin are used. These could develop AMR and pass to human pathogens. This in turn could endanger effective clinical treatment of bacterial infections in patients (Cabello, 2006). The antibiotics like oxytetracycline, sulfamerazin, flumequine, sarafloxacin, erythromycin and ormethoprim are used in large water bodies for safe guarding farmed fish (Chanda et al., 2012). Fish farms widely use tetracycline and chloramphenicol, which

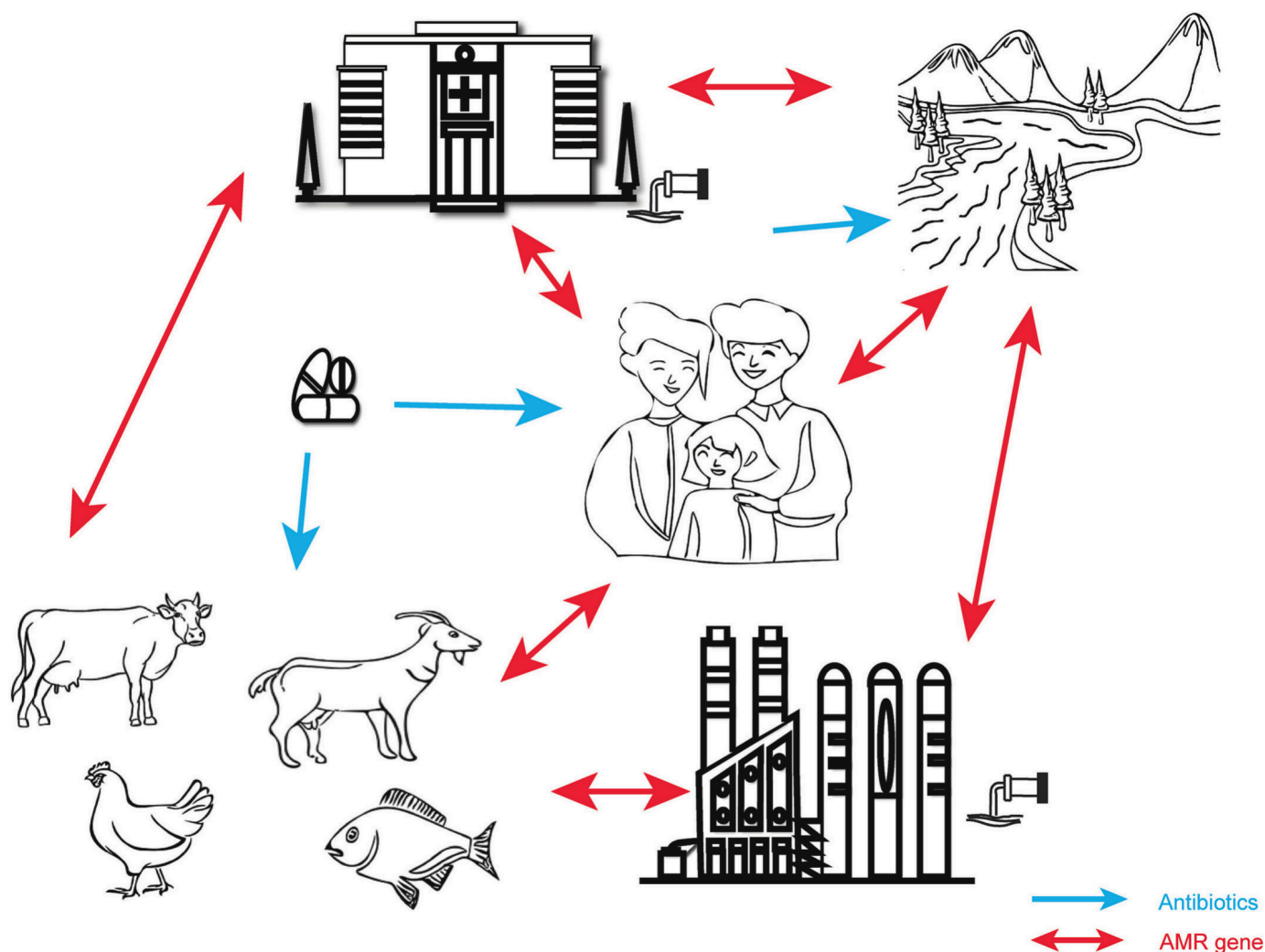


Fig. 1. Antimicrobial resistance element circulation routes in the food chain The antimicrobials or antibiotics are used to treat diseases in human, animals and in agriculture. Under the sustained or indiscriminate use of antibiotics the target microbes evolve ways to overcome deleterious effect of antibiotics and develop resistance. Such resistant genetic elements are transmitted horizontally or vertically to other organisms or the components in the food chain and gets amplified. This vicious cycle continues.

are the common antibiotics used in clinics too (Marshall and Levy, 2011). Therefore AMR emergence in fish could pose risk of limiting the use of same antibiotics in clinic. However, ground data from fisheries industry are not always available on AMU and AMR patterns, especially from LMICs like India (Manyi-Loh et al., 2018). Hence AMR surveillance in fish is a necessary step to understand emerging risk and for its assessment.

In India, 55 % of the total fish production is from freshwater aquaculture. A survey of freshwater fish hatcheries in West Bengal from 2006 to 2007 noted that althrocin, ampicillin, enrofloxacin, oxytetracycline and sparflaxacin were commonly used for the prophylaxis and treatment (Abraham and Bharathkumar, 2011). The southern states of India are predicated to be the hotspots of multidrug resistance (Schar et al., 2021), which is in agreement with a previous report (Deekshit et al., 2012). In view of the above matters, the Coastal Aquaculture Authority of India has recently banned the use of twenty antibiotics or pharmacologically active substances, including chloramphenicol, especially use in shrimp aquaculture (Coastal Aquaculture Authority et al., 2022).

4.2. Poultry

Poultry is domesticated birds raised for meat, eggs, manure, fiber, feathers etc. They can be also a flock used for entertainment. Poultry is considered fastest growing livestock used in different countries for lean protein food. Brazil, China, the European Union, and the USA are leading poultry producers (Hedman et al., 2020). Poultry farming gained impetus worldwide because it is easy to manage owing to many features like their portable small body size, short life span, and high efficiency energy absorption and quick adaptability to rough environmental conditions unlike other terrestrial livestock (Hedman et al., 2020). Poultry farms routinely use antimicrobials for intensive farming practices. The use of antimicrobials in poultry was somewhat accidental (Moore et al., 1946) and the earliest use of antimicrobial growth promoters (AGPs) was in chickens. Arsenical compounds, bacitracin, flavophospholipol and virginiamycin are some AGPs used in poultry production. The AGPs like avoparcin (banned in EU), bacitracin and virginiamycin are used to control gram positive bacteria like *Clostridium perfringens*, which is commonly causing food poisoning and is potentially fatal (Wegener et al., 1999). Initially, antimicrobials were used for growth promotion and disease prevention but later on it became a major component in the intensive growth of poultry production; its use quickly spread to LMICs. The use of antimicrobials in poultry was reported by World Organization for Animal Health, OIE in 2016. The data were collected from 93 countries and 87.13 % of antibiotics used were reportedly from America, Asia, and East Oceania. Tetracycline and penicillin were used the most (de Mesquita Souza Saraiva et al., 2021).

The United States Food and Drug Administration (FDA) authorized use of antimicrobials in feed without the veterinary prescription in 1951. The practice of antimicrobial use in the feed was varied among various countries. In 2006 the European Union Regulation (No.1831/2003) limited the use of antimicrobials for animal nutrition, i.e. use of antimicrobials in the feed which are beyond the treatment of coccidiosis and histomonosistats (Hedman et al., 2020). After that in 2013 the FDA restricted use of AGPs, which are important medicines for humans. In some countries based on FDA restrictions AGPs are banned but the non-OECD (Organization for Economic Cooperation and Development) countries like Brazil, India, South Africa etc. are considered the leading poultry producers still use AGPs (Hedman et al., 2020). The effect of banning of AGPs in economy will be less in high income countries compared with low-income countries with poor biosecurity and sanitation practices. The reduction in use of antimicrobials gives chance for disease in animal population, which is maintained in poor sanitary conditions. On the other hand, unregulated use of antibiotics leads to AMR, as seen in LMICs (Hedman et al., 2020).

The leaping AMR trend in poultry may lead to treatment failure and economic losses. The development of resistant genes or organisms may

cause risk to the human population. The common pathogens in poultry are *Escherichia coli*, *Clostridium perfringens*, *Mycoplasma* spp., *Salmonella* Pullorum/Gallinarum, *Pasteurella multocida* etc. Among them, avian pathogenic *E. coli* (APEC) shows higher level of antimicrobial resistance against various antibiotics. In Africa, Asia, Brazil, Spain APEC have >70 % resistance for ampicillin but in the United States it was 40 % in 2001–2003 (Nhung et al., 2017). The resistance for tetracycline in APEC was between 40 % and 100 % in these countries. In Jordan resistance to amoxicillin in APEC was 100 % and in Brazil it was 74 % (Nhung et al., 2017). It causes colibacillosis in poultry. *Ornithobacterium rhinotracheale*, is a gram-negative bacterium that shows highest phenotypic resistance with median AMR >50 % to co-trimoxazole, enrofloxacin, gentamicin, amoxicillin, and ceftiofur (Nhung et al., 2017). It causes respiratory infections in chickens, turkeys and other avian species. It shows resistance towards different antimicrobials (Nhung et al., 2017). *Pasteurella multocida* shows a lesser AMR rate i.e. <20 % against all antimicrobials (Nhung et al., 2017). It causes fowl cholera which leads to acute fatal septicemia in adult birds. A report from India showed that it is 100 % resistant to sulfadiazine, a common drug used to treat fowl cholera (Nhung et al., 2017).

Salmonella Pullorum/Gallinarum cause septicemic diseases in poultry called pullorum disease and fowl typhoid, respectively (Nhung et al., 2017). *Avibacterium paragallinarum* is another pathogen which causes respiratory disease in chickens. All these pathogenic organisms are resistant towards ampicillin, amoxicillin, ceftiofur, streptomycin, gentamicin, kanamycin, chloramphenicol, florfenicol, tetracycline, co-trimoxazole, nalidixic acid, ciprofloxacin and enrofloxacin (Nhung et al., 2017).

Poultry is a common source harboring multi-resistant bacteria. It could be readily monitored for AMR gene spread in the ecosystem. Though many countries have good veterinary services most do not meet the international standards. A proper veterinarian service can improve the diagnosis and treatment of various diseases related to poultry without leading to AMR by use of prescription antimicrobials (Hedman et al., 2020).

4.3. Cattle

Milk is a good source of many essential nutrients and is consumed by all age groups. The last three decades witnessed profound increase in world milk production i.e. ~60 % (FAO, 2022). South Asia is the major contributor of milk production in the developing countries since 1970. India contributes largest share (22 %) of world milk production. The population growth, increasing in income and enhanced health awareness have driven high demand for dairy products across the world (FAO, 2022). This is reflected in the increased momentum for cattle farming. Australia, France, Germany, Ireland, New Zealand and the United States of America produce surplus amount of milk whereas Algeria, China and Indonesia, Italy, Mexico, the Russian Federation have deficits in milk production.

Mastitis remains a major problem faced by the farmers of dairy farms. Many factors are responsible for mastitis such as the unhygienic practice of milking, failure to disinfect cow sheds, and lack of care in isolating sick cows and buffaloes (Joshi and Gokhale, 2006). Milk produced by mastitic cows and buffaloes has been shown to have bacteria with AMR against commonly used antibiotics in dairy farms (Bandyopadhyay et al., 2014). Methicillin resistant *S. aureus* is a growing concern for dairy farmers (Shah et al., 2019). Antibiotic resistant *E. coli* has been detected in dairy cattle back in 2005 (Sawant et al., 2005). Farmers also add antibiotics to feed and water, they retain the bacterial strains which are antibiotic resistant and may proliferate within the animal population. It can also transmit to other animals and farmers and workers of slaughterhouses and ultimately to common man (Fig. 1).

Similarly, in the past 50 years global demand for meat is growing and meat production has tripled. Now the world produces >340 million tonnes annually. At present Asia is the largest meat producer (45 %). In

1961, Asia produced only 12 % whereas Europe and North America were the dominant meat producers. By 2013, Europe and North America's share had declined drastically (Data, 2018).

India is the top producer of buffalo meat and is the second in goat meat production, and is the largest exporter of sheep and goat meat (Government of India, Dept. Animal Husbandry and Dairying, 2021). To meet the high demand of yield maintaining the health of animals and limit the mortality and morbidity antimicrobials have been heavily used in cattle (Sharma et al., 2018). Amoxicillin, ampicillin, cloxacillin, gentamicin oxytetracycline, tetracycline and penicillin are most frequently used in cattle owing to lower costs (Kumar et al., 2021). Shiga toxin producing *E. coli* with AMR has been isolated from calves suffering from diarrhea (Arya et al., 2008). Non-typhoid *Salmonella* with AMR was reported in meat from Vietnam (Nhung et al., 2018) and Switzerland (Collineau et al., 2017). An *Enterococcus* spp. harboring AMR was isolated from a meat processing unit and in the ground meat from retail market of Canada (Holman et al., 2021). This is an indicator of how far the risk of AMR in food chain has evolved.

4.4. Pigs

Pigs are one of the most widely reared livestock in the world for meat. They are the basic source of protein for millions of individuals around the globe, especially in Europe and Southeast Asia. Pork has low fat content and its texture and palatability quickly attract the consumers, and most importantly it is cheaper than other meat. Therefore it is sought after meat product. China hosts half of the pig population of world followed by the EU and the USA (FAO data). In 2022, 780 million pigs were globally available and it is an increase of 30 million from previous year (Shahbandeh, 2022). Definitely the consumption of it is rapidly increasing in low income settings. In 2021, the global pork production had increased to 105 million metric tonnes and China was the leading country in the market (Mezoughem, 2022). In North-Eastern India pig farming is considered one of the main profitable livestock businesses. India has a total of 9.06 million pig heads and 46.88 % of them are distributed among eight North-Eastern states. Assam has the highest pig population in India followed by Jharkhand, Meghalaya, and West Bengal (Government of India, Dept. Animal Husbandry and Dairying, 2021, 2021). Pig farming often encounters communicable diseases and antibiotics use is inevitable for reducing morbidity and mortality. Surprisingly an estimate suggested that pig farms consume largest quantity of antimicrobials among livestock (Van Boeckel et al., 2015). The pig farms of North Eastern states of India use amoxicillin, ampicillin, cephalixin cephalosporin, penicillin, and plus clavulanic acid are commonly used antibiotics for therapeutic purposes. Monensin, amprolium, bacitracin, neomycin, and chlortetracycline were the antibiotics used in the feed for prophylaxis. Antibiotics used for both therapeutic and prophylaxis purposes are amoxicillin, oxytetracycline, and streptomycin (Satish Kumar et al., 2020). Alarming the North East Region of India is an established hotspot for AMR bacteria (Van Boeckel et al., 2019).

5. Nosocomial settings as an AMR reservoir

The patients often acquire infections from the hospital and 15 % of patients globally were susceptible for nosocomial infection. The hospital environment is highly prone for microbial contamination and the chance for presence of AMR microbes (D'Accolti et al., 2019). Poor sanitization practices and improper discharge of bio-waste are some of the reasons for AMR. The excessive usage of prescription antibiotics in patients aggravated AMR crisis. Every year in Europe, 3 million patients have been diagnosed with nosocomial infection and about 37,000 patients die from its consequence. The main pathogens with AMR that caused 80 % of the deaths in a descending order are *Escherichia coli*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Streptococcus pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa* (Christopher et al., 2022). In Italy,

nosocomial infection has become more common and the death rate has rapidly increased (D'Accolti et al., 2019). The patients with burns, in ICUs, undergoing organ transplant and infants are the main victims of nosocomial infection. In Serbia, a recent study reported that one in three patients admitted to ICU was suffering of nosocomial infection. In neonates of Sub-Saharan Africa and South-East Asia, nosocomial infection was the main reason for three out of four lethal cases (WHO, 2011). In 2016, Pakistan health authorities reported an outbreak of typhoid fever which was extensively drug-resistant (XDR). About 5274 cases have been identified in 2018 and were transmitted worldwide (Jacobs et al., 2019). About 2,00,000 nosocomial infection cases including *Clostridium difficile* infection and catheter-associated urinary tract infections occur in Australia every year and about 7000 deaths were annually reported (Fernando et al., 2017). In Greece, a recent study reported that mortality risk doubled in patients with nosocomial infection of carbapenem-resistant pathogens compared to patients without infection (Avershina et al., 2021). The weakness in surveillance systems for detecting nosocomial infection exists in high income countries like the USA, the Netherlands, Switzerland and Japan (Greene et al., 2021). It suggests, the scenario may be much worse in LMICs.

India reported the emerging AMR in tertiary care settings, where *Escherichia coli* is the top ranking, followed by *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Acinetobacter baumannii* and *Staphylococcus aureus* (ICMR, 2020). *Staphylococcus aureus* has increasing trend of acquiring AMR as it displays resistance against most antibiotics. Fungal infections of rare species are increasing in hospitalized patients and its treatment demands novel strategies. *Candida auris* infections are common in Indian hospitals. *C. auris* is multi-drug resistant and its resistance against fluconazole and echinocandin is on the rise. The scenario of AMR in primary and secondary level health care is unclear. Currently, three groups of pathogens are recognized: Group I: high risk pathogens, which developed resistance to last resort of antibiotics like carbapenems that is used for multi-drug resistant bacteria. Group II: high risk multi-drug resistant pathogens. Both Group I and II pathogens are acquired in hospitals, like ventilator associated pneumonia, urinary tract infections and infection in site of surgery. Group III: are drug resistant pathogens like *Enterobacterales* and needs careful surveillance. The group I and II cause higher rates of mortality and morbidity in India as is the case in global scenario too. The hospital with poor incineration practice has high chance of spilling AMR microbes into environment. This is a reality because the clinical experience testifies that certain pathogens are becoming highly resistant to antibiotics. To assess this risk primary and secondary hospitals must be proactively monitored.

6. One Health approach to limit emerging threat of AMR

The emerging AMR is slowly depleting the reserve antibiotics used in clinics. It is a serious health care challenge because the pace of new antibiotic development is not matching with speed of developing the antibiotic resistance (Hutchings et al., 2019). The WHO recognized this rapidly emerging risk of AMR worldwide and passed a resolution in World Health Assembly on 28 May 2019 (WHO, 2019). It calls for increased efforts by all the member states for enhancing a multi-prong approach to develop innovative tools that can effectively reduce antimicrobial resistance. It advocates for a One Health approach integrating all sectors where antibiotics are used and/or permeated. The coordinated and responsible efforts are required to achieve the sustainable health care goals like curbing AMR.

The menace of self-medication is a deterrent to the goal of One Health approach for minimizing antibiotics use. In fact, antibiotics are rampantly used in humans and animals without prescription in many countries, especially in LMICs (Nepal and Bhatta, 2018; Svenson et al., 2021)34803044, 29876150. The self-medication is more in rural and semi-urban settings perhaps due to less access to health care providers (Svenson et al., 2021) 34803044. However, the irrational self-medication is very dangerous because without proper diagnosis of the

microorganism causing an illness antibiotic consuming may not serve its intended purpose instead give opportunity for even commensals in gut to develop AMR. For example amoxicillin is self-medicated the most in humans and animals (Nepal and Bhatta, 2018) 29876150, and it can lead *E. coli* in gut to develop AMR. In many countries antibiotics can be easily purchased without a prescription (Svenson et al., 2021) 34803044. This would certainly amplify AMR emergence in local settings. Therefore, diagnosis based prescription must be enforced for antibiotics use in humans and animals. A system of surveillance for AMU must be strictly implemented in all the countries because AMR spreads beyond the borders. The production and consumption of antibiotics should be monitored and must be judiciously audited to check any unauthorized use in the communities. This will help weed out self-medication, an overlooked loophole for AMR development in the food chain.

Innovative approaches in research and development are required to ensure quality, safety and efficacy of antimicrobials. Additionally, affordability of antibiotics is also a concern. Another avenue for research is to find natural alternatives for antibiotics in order to reduce their use. Antibiotics may be replaced with vaccines. It will be a viable alternative for at least few of the pathogens. For example, *Pneumococci* and rotavirus vaccines prevented 23.8 and 13.6 million of infections, respectively in LMICs (Lewnard et al., 2020). One Health approach is meant to reduce AMR in a comprehensive manner from the environment, plants, animals and humans. The weakest link in a food chain unattended can recharge back AMR to the food chain and it could defeat the aim of One Health. Therefore, a coordinated effort in every sector is compulsory to limit the transmission of AMR.

The unsparing use of antibiotics by humans, and indiscriminate use of it in clinics, livestock and agriculture for decades resulted in AMR development against many antibiotics. The past effort to trace the sources of AMR was largely focused on studies involving human infections, and how antibiotics respond. The new trend in certain countries is that antibiotics are more used in livestock than in humans (Van Boeckel et al., 2019); and lately they are also used for agriculture (Reeder, 2020). Thus, biotic sources like animals and plants, and abiotic components of environment are all source of AMR resistome (Facciola et al., 2021). The WHO, FAO, OIE, and the UN Environment Programme (UNEP) have jointly introduced the quadripartite agreement on the 'One Health approach' to tackle the risk of AMR emerging from all these sources in environment (FAO et al., 2022).

6.1. Steps to stop AMR

The first step to stop spread of AMR is to make aware of its risk to all the parties involved. Many preliminary steps are necessary before the public is made aware of AMR. Some of them are: 1). Identification of ecological niches and routes of transmission of AMR in the food chain (Acar and Moulin, 2006). 2). Understand the public health implications of AMR in human population/communities, especially when niches are intermingled (Thanner et al., 2016). Addressing first these issues will enable the local administration and organization to educate communities, who are living in a niche with AMR risk.

6.2. Role of governmental organizations in controlling the use of antimicrobial agents in food

The WHO proposed the One Health approach to limit threat of antibiotic resistance. The "Tripartite Alliance of the WHO, FAO, and OIE published the Global Action Plan on Antimicrobial Resistance in 2015 and it is an attempt to counteract effectively the worldwide concern of AMR (WHO, 2015). The FAO likewise released an Antimicrobial Resistance Strategy in September 2016 for better agricultural practices to adopt.

The five strategic objectives of the WHO's Global Action Plan are:

- Heighten awareness and understanding on antibiotic use and antibiotic resistance
- Strengthen knowledge via surveillance and research
- Reduce infectious diseases
- Optimize rational antibiotic use
- Mobilize resources for research and develop framework to set out integrated prevention and containment measures of anti-microbial resistance in the food chain (WHO, 2015).

To achieve above five goals the policy makers must be properly educated because they are a key stone of the whole system in every administrative step of government. Their understanding of the One Health approach will dictate successful implementation of it on ground. Therefore, they must be oriented on the scientific rationale for One Health Approach with continuous education, on regular basis. They should be aware of what is evolving on the ground regarding AMR and the rational methods available to control the emerging threat. They must be minimum sensitized on what One Health approach is and why and how to implement it. This can be achieved by constant exchange of information between the scientific community and the policy makers. An annual appraisal of AMR status from ground level to top policy makers is a must if we are serious and want to fully address it in the frame work of One Health approach.

6.3. Need to map AMR and stop rapid depletion of antibiotics and identification of potential microorganisms which are capable of causing epidemic like situations

Research should be carried out for creating a database of AMR microbes found in particular region/niche. The information can be given to policy makers in the government and regulatory bodies, which in turn will be able to educate medical practitioners, local farmers and food handlers for better practices. In parallel, increasing the pace of research and development for novel antibiotics is inevitable as the repertoire of antibiotics are depleting faster than the spread of AMR.

6.4. Role of scientific organizations in finding solutions against AMR strains

The microbes develop AMR as a mechanism of survival in the environment. It is achieved through two biological process.

- Resistance elements can be inherited from pre-existing strains in population.
- Spontaneous genetic variation arises in microbes as acquired resistance, which could vertically or horizontally transfer to its progeny or other organism, respectively.

The niche specific selection pressure for AMR emergence is not understood. It requires more research to understand under what circumstances AMR genes are emerging. Obviously the constant and enhanced load of antibiotics encountered by bacteria primarily creates selection pressure on them. Hence, best option is to reduce use of antibiotics itself to bring down AMR. In addition, identifying the vulnerable factors inducing AMR could help map the susceptibility in a niche. Based on such information, the scientific community can educate local authorities to take precautionary actions. Scientific communities knew that vaccines should be used more instead of antibiotics to reduce AMR. This idea should be appreciated and practiced where ever possible. Furthermore, encouragement and funding should be allocated for development of vaccines against pathogens posing AMR risk. However, it should be kept in mind that developing an effective vaccine against each and every microbes may not be simply possible. Recent discoveries on adjuvants can be taken into account for development of new drugs.

6.5. Policy guidelines of WHO for curbing AMR

The 71st General Assembly of UN on 13th September 2016 generated a global plan for the member states of the WHO to collect and report antibiotic consumption data. It mentioned the critically important antimicrobials that are in use in human medicine include quinolones, cephalosporins, macrolides and ketolides, glycopeptides, and polymyxins. Emergence of AMR in these classes of antibiotics will be of serious consequence to the health management of patients. In 2021, WHO adopted a new guideline (FAO, 2021) on integrated monitoring and surveillance of foodborne antimicrobial resistance to the emergence of AMR. It highlighted need to monitor the foodborne pathogens causing public health risk and indicator bacteria. It requires innovative design and implementation of integrated monitoring and surveillance systems for AMR and AMU in place all along the food production and supply chains. The guidelines put forward nine basic principles. The foremost one is to apply One Health approach whenever possible and applicable to monitor and surveillance of foodborne AMR. This should be part of national strategies and policy to reduce AMR. The risk analysis should be a criterion to design and implement surveillance programs. The programmers should generate data on AMR and AMU in relevant domains and give input on risk analysis. The programs should be designed to meet national priorities and improve quality within the resource limit. It should prioritize the monitoring of most relevant foodborne AMR and AMU safety issues of public health concern and of national importance. It should focus on new emerging AMR pathogens and assess epidemiological risk involved. The monitoring laboratories must implement effective quality assurance and management systems. The quality of methods for monitoring and surveillance must be of international standards and verifiable methods. To manage AMR monitoring and surveillance capacity building with latest technology, like whole genome sequencing of bacteria is required, and training of new manpower in these techniques are also essential. AMR trend can dynamically change with climate and time, which necessitates recurrent analysis of AMR.

6.6. Modern methods to rapidly monitor emerging AMR genes

Identification of a bacterium was classically performed by culturing it in the laboratory. However, many bacteria are fastidious and cannot be cultured in the laboratory conditions. It limits the ability to identify and characterize them. The phylogenetic analysis is the alternative for this bottleneck. To quickly identify the phylogeny of bacteria in a sample, a 16S sequencing can be performed. It provides information of the diversity of bacteria based on a single ribosomal gene, *16S rRNA*. The *16S rRNA* has nine variable regions flanked by conserved stretches in the majority of bacteria (Johnson et al., 2019). The conserved regions are used as the targets for PCR primers. The hypervariable region sequences provide the species-specific signature for identification of the bacteria though with limitations to discriminate bacterial species and strains (Johnson et al., 2019). However, this method is technically unsuitable for gathering information to identify the emerging AMR genes in bacteria, which requires shotgun metagenomics. Shotgun metagenomics is capable of simultaneously identifying thousands of organisms in a single sample by next generation parallel sequencing. It scans all the genes in a bacterium, which permits bioinformatics tools to identify functional genes by interrogating against the known AMR databases like ResFinder and National Database of Antibiotics Resistant Organisms. Thus shotgun metagenomics is a very powerful method to obtain a comprehensive map of emerging AMR genes in a niche and to provide insight into the bacterial community biodiversity (Munk et al., 2018). This method could be adapted to detect low abundance bacterial strains in a community.

7. Challenges and future directions

The AMR is a serious health risk emerging worldwide. The indiscreet use of antibiotics for treatment, for agricultural crop improvement and food processing by practice of modern methods has enhanced occurrence and transmission of AMR through food chain. Food chain is the major source of AMR in LMICs because of unrestricted availability and indiscriminate use of antibiotics. More interestingly, AMR is increasingly a challenging issue in emerging economies because their purchasing power and consumption rate have enormously increased. One of the direct impacts of this is increased consumption of meat and egg by the general population that is a major impetus for increased production of livestock like cow and poultry. This has been achieved by rampant use of antibiotics in livestock. In addition, the increased consumption of meat in LMICs proportionately increased AMR (Manyi-Loh et al., 2018). Similarly, to increase food security the increased production of crops is necessary, which also depends on use of antibiotics and other pest management chemicals. This trend was earlier seen in EU and the USA, which was intervened by new policy decisions. A similar decision by governments in LMICs is required to curb rapid spreading of AMR. The clinicians, pharmacies, vendors and farmers must be educated for limited use of antibiotics to check pervading threat of AMR before it gets out of control.

The whole resistome can be assessed with new generation sequencing technology. It could help assess total AMR burden in a niche (Walsh, 2018). This assessment is possible based on the available information about the AMR resistome. However, currently the databases are incomplete with information, which is a limitation. Resistome may be much larger than what is perceived based on the available data. Discovery of *mobilized colistin resistance (mcr)* gene revealed that many other databases had *mcr* gene (Liu et al., 2015). While assessing the AMR climatic and anthropogenic effects have to be accounted to get accurate measures. The prevalence of AMR must be correlated with the epidemiological data inputs. One health approach has many challenges. The main one are: 1). Sampling issue: To get meaningful information the proper and random sampling are necessary. Otherwise the information obtained may be largely skewed and it might provide an under or overestimated perception of the threat. 2). To measure AMR prevalence in space is still challenging because it is a dynamically changing domain. 3). It is important to detect phenotypic and genotypic resistance and is necessary for capturing phantom resistance. Therefore, functional resistome data must be generated and it needs metagenomics approaches to sample large area and more number. It could help prioritize rational use of antibiotics. 4). One of the most challenging issues is to factor climatic and anthropogenic effects on AMR. The changing weather, natural calamities, pandemic like COVID-19 could drastically change the ground realities and conventional methods may not accurately capture and assess such conditions (Hirabayashi et al., 2021). Anthropogenic changes like huge industrial effluents, hospital waste etc. could lopsided local niche in communities. While assessing such conditions proper attention should be given to various confounding factors. 5). It is very important to correlate the prevalence and epidemiological data with the data obtained from the source. A real AMR threat should be pervading in the locality within various component of the food chain. Thus a realistic picture of the AMR should be obtained using various tools before pressing the panic button and shutting of production houses, the life sustaining systems of communities. All these could be easily addressed in small scale studies very systematically (Wang et al., 2017) but scaling up of the volume could introduce significant experimental errors. The development in artificial intelligence and big data analysis may help to address such complex system level challenges present in the food chain.

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SV conceived the review, PS and AA wrote the initial draft and developed the figure. SKB contributed to the AMR policy section and edited the draft. SV edited the final draft.

Declaration of competing interest

The authors have no competing interests to declare.

Data availability

No data was used for the research described in the article.

References

- Abraham, T.J., Bharathkumar, G., 2011. Distribution of Motile Aeromonads, Pseudomonads and Oxytetracycline Resistant Bacteria in Freshwater Catfish *Pangasius pangasius* Hatcheries of West Bengal, India. *J. Biosci.* 17, 13–20.
- Acar, J.F., Moulin, G., 2006. Antimicrobial resistance at farm level. *Rev. Sci. Tech.* 25 (2), 775–792.
- Adesogana, Adegbola T., McKunec, Sarah L., Eilittad, Marjatta, Dahl, G.E., AHH, 2020. Animal source foods: Sustainability problem or malnutrition and sustainability solution? Perspective matters. *Global Food Security* 25 (2020), 100325.
- Arino, A., Beltran, J.A., Herrera, A., Roncales, P., 2013. Fish and seafood: Nutritional value. In: Allen, L.H., Prentice, A., Caballero, B. (Eds.), *Encyclopedia of Human Nutrition*, Third Edition. Academic Press, London, ISBN 978-0-12-384885-7, pp. 254–261.
- Arya, G., Roy, A., Choudhary, V., Yadav, M.M., Joshi, C.G., 2008. Serogroups, atypical biochemical characters, colicinogeny and antibiotic resistance pattern of Shiga toxin-producing *Escherichia coli* isolated from diarrhoeic calves in Gujarat, India. *Zoonoses Public Health* 55 (2), 89–98.
- Avershina, E., Shapovalova, V., Shipulin, G., 2021. Fighting antibiotic resistance in hospital-acquired infections: current state and emerging technologies in disease prevention, diagnostics and therapy. *Front. Microbiol.* 12, 707330.
- Baker-Austin, C., Dopson, M., 2007. Life in acid: pH homeostasis in acidophiles. *Trends Microbiol.* 15 (4), 165–171.
- Bandyopadhyay, S., Samanta, I., Bhattacharyya, D., Nanda, P.K., Kar, D., Chowdhury, J., Dandapat, P., Das, A.K., Batul, N., Mondal, B., Dutta, T.K., Das, G., Das, B.C., Naskar, S., Bandyopadhyay, U.K., Das, S.C., 2014. Co-infection of methicillin-resistant *Staphylococcus epidermidis*, methicillin-resistant *Staphylococcus aureus* and extended spectrum beta-lactamase producing *Escherichia coli* in bovine mastitis—three cases reported from India. *Vet Q* 35 (1), 56–61.
- Bar-On, Y.M., Phillips, R., Milo, R., 2018. The biomass distribution on earth. *Proc. Natl. Acad. Sci. U. S. A.* 115 (25), 6506–6511.
- Cabello, F.C., 2006. Heavy use of prophylactic antibiotics in aquaculture: a growing problem for human and animal health and for the environment. *Environ. Microbiol.* 8 (7), 1137–1144.
- Cassini, A., Hogberg, L.D., Plachouras, D., Quattrocchi, A., Hoxha, A., Simonsen, G.S., Colomb-Cotinat, M., Kretzschmar, M.E., Devleeschauwer, B., Cecchini, M., Ouakrim, D.A., Oliveira, T.C., Struelens, M.J., Suetens, C., Monnet, D.L., 2018. Attributable deaths and disability-adjusted life-years caused by infections with antibiotic-resistant bacteria in the EU and the European economic area in 2015: a population-level modelling analysis. *Lancet Infect. Dis.* 19 (1), 56–66.
- Chanda, M., Paul, M., Maity, J., Dash, G., Gupta, S.S., 2012. The use of antibiotics and disinfectants in ornamental fish farms of West Bengal, India. *J. Nat. Sci. Biol. Med.* 2 (2), 139–140.
- Christopher, J.L., Murray, C.J.L.M., Ikuta, Kevin Shunji, Sharara, Fablina, Lucien Swetschinski, G.R.A., Gray, Authia, Han, Chieh, Bisignano, C., 2022. Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis. *Lancet* 399 (10325), 629–655.
- Coastal Aquaculture Authority, M.O.F., Dept of Fisheries, Government of India, Animal Husbandry and Dairying, 2022. List of Antibiotics and Other Pharmacologically Active Substances Banned for Use in Shrimp Aquaculture. Departmental Order 33035/4/2002/-Fy (T-2)/(Trade) (E-17739).
- Collignon, P.J., Conly, J.M., Andremont, A., McEwen, S.A., Aidara-Kane, A., Agero, Y., Collignon, P., Conly, J., Dang Ninh, T., Donado-Godoy, P., Fedorka-Cray, P., Fernandez, H., Galas, M., Irwin, R., Karp, B., Matar, G., McDermott, P., McEwen, S., Mitema, E., Reid-Smith, R., Scott, H.M., Singh, R., DeWaal, C.S., Stelling, J., Toleman, M., Watanabe, H., Woo, G.J., 2016. World Health Organization ranking of antimicrobials according to their importance in human medicine: a critical step for developing risk management strategies to control antimicrobial resistance from food animal production. *Clin. Infect. Dis.* 63 (8), 1087–1093.
- Collineau, L., Carmo, L.P., Endimiani, A., Magouras, I., Muntener, C., Schupbach-Regula, G., Stark, K.D.C., 2017. Risk ranking of antimicrobial-resistant hazards found in meat in Switzerland. *Risk Anal.* 38 (5), 1070–1084.
- D'Accolti, M., Soffritti, I., Mazzacane, S., Caselli, E., 2019. Fighting AMR in the healthcare environment: microbiome-based sanitation approaches and monitoring tools. *Int. J. Mol. Sci.* 20 (7).
- Data, S.A., 2018. Meat Production by Country- 1961 to 2018.
- de Mesquita Souza Saraiva, M., Lim, K., do Monte, D.F.M., Givisiez, P.E.N., Alves, L.B.R., de Freitas Neto, O.C., Kariuki, S., Junior, A.B., de Oliveira, C.J.B., Gebreyes, W.A., 2021. Antimicrobial resistance in the globalized food chain: a One Health perspective applied to the poultry industry. *Braz. J. Microbiol.* 53 (1), 465–486.
- Deekshit, V.K., Kumar, B.K., Rai, P., Srikanth, S., Karunasagar, I., Karunasagar, I., 2012. Detection of class 1 integrons in salmonella weltevreden and silent antibiotic resistance genes in some seafood-associated nontyphoidal isolates of salmonella in south-west coast of India. *J. Appl. Microbiol.* 112 (6), 1113–1122.
- ECDC, 2022. Total Care Antimicrobial Consumption (Community and Hospital Sector) for Year 2021.
- ECDC, E.a., 2016. The European Union summary report on antimicrobial resistance in zoonotic and in dicatobacteria from humans, animals and food in 2014. *EFSAJ.* 14, 438.
- EFSA (European Food Safety Authority), 2014. Annual Report of the EFSA Journal 2013. EFSA Supporting publication. EN-721. 9pp.
- EMA/795956/2022, 2022. Sales of Veterinary Antimicrobial Agents in 31 European Countries in 2021, 2022. European Medicines Agency, European Surveillance of Veterinary Antimicrobial Consumption.
- Facciola, A., Lagana, P., Caruso, G., 2021. The COVID-19 pandemic and its implications on the environment. *Environ. Res.* 201, 111648.
- FAO, 2022. Gateway to Dairy Production and Products.
- FAO, WHO, 2019. Codex 2019: The Year of Food Safety.
- FAO, O., UNEP, WHO, 2022. Quadripartite Memorandum of Understanding (MoU) Signed for a New Era of One Health Collaboration.
- FAO, T., 2021. Guidelines on integrated monitoring and surveillance of foodborne antimicrobial resistance. In: *The Role of Codex in AMR - CXG 94-2021*.
- Fernando, S.A., Gray, T.J., Gottlieb, T., 2017. Healthcare-acquired infections: prevention strategies. *Intern. Med. J.* 47 (12), 1341–1351.
- Fleming, A., 2001. On the antibacterial action of cultures of a penicillium, with special reference to their use in the isolation of *B. influenzae*. 1929. *Bull World Health Organ* 79 (8), 780–790.
- Flemming, H.C., Wuertz, S., 2019. Bacteria and archaea on earth and their abundance in biofilms. *Nat. Rev. Microbiol.* 17 (4), 247–260.
- Fotuhi, M., Mohassel, P., Yaffe, K., 2009. Fish consumption, long-chain omega-3 fatty acids and risk of cognitive decline or Alzheimer disease: a complex association. *Nat. Clin. Pract. Neurol.* 5 (3), 140–152.
- Government of India, Dept. Animal Husbandry and Dairying, 2021. Livestock Production in India.
- Greene, M.T., Krein, S.L., Huis, A., Hulscher, M., Sax, H., Sakamoto, F., Sakihama, T., Tokuda, Y., Fowler, K.E., Saint, S., 2021. Infection prevention practices in the United States, the Netherlands, Switzerland, and Japan: results from national surveys. *Infect. Control Hosp. Epidemiol.* 42 (10), 1206–1214.
- Guillen, J., Natale, F., Carvalho, N., Casey, J., Hofherr, J., Druon, J.N., Fiore, G., Gibin, M., Zanzi, A., Martinsohn, J.T., 2018. Global seafood consumption footprint. *Ambio* 48 (2), 111–122.
- Gusberti, M., Klemm, U., Meier, M.S., Maurhofer, M., Hunger-Glaser, I., 2015. Fire blight control: the struggle goes on. A comparison of different fire blight control methods in Switzerland with respect to biosafety, efficacy and durability. *Int. J. Environ. Res. Public Health* 12 (9), 11422–11447.
- Hedman, H.D., Vasco, K.A., Zhang, L., 2020. A review of antimicrobial resistance in poultry farming within low-resource settings. *Animals (Basel)* 10 (8).
- Hesp, A., Veldman, K., van der Goot, J., Mevius, D., van Schaik, G., 2019. Monitoring antimicrobial resistance trends in commensal *Escherichia coli* from livestock, the Netherlands, 1998 to 2016. *Euro Surveill.* 24 (25).
- Hirabayashi, A., Kajihara, T., Yahara, K., Shibayama, K., Sugai, M., 2021. Impact of the COVID-19 pandemic on the surveillance of antimicrobial resistance. *J. Hosp. Infect.* 117, 147–156.
- Hodgkin, D.C., 1949. The X-ray analysis of the structure of penicillin. *Adv. Sci.* 6 (22), 85–89.
- Holman, D.B., Klima, C.L., Gzyl, K.E., Zaheer, R., Service, C., Jones, T.H., McAllister, T. A., 2021. Antimicrobial resistance in *Enterococcus* spp. isolated from a beef processing plant and retail ground beef. *Microbiol Spectr* 9 (3), e0198021.
- Hutchings, M.L., Truman, A.W., Wilkinson, B., 2019. Antibiotics: past, present and future. *Curr. Opin. Microbiol.* 51, 72–80.
- ICMR, 2020. Antimicrobial Resistance Research and Surveillance Network, January to December 2020. ICMR Annual Report. Division of Epidemiology and Communicable Diseases, ICMR, New Delhi.
- Jacobs, J., Hardy, L., Semret, M., Lunguya, O., Phe, T., Affolabi, D., Yansouni, C., Vandenberg, O., 2019. Diagnostic bacteriology in district hospitals in sub-saharan Africa: at the forefront of the containment of antimicrobial resistance. *Front. Med (Lausanne)* 6, 205.
- Jans, C., Sarno, E., Collineau, L., Meile, L., Stark, K.D.C., Stephan, R., 2018. Consumer exposure to antimicrobial resistant bacteria from food at Swiss retail level. *Front. Microbiol.* 9, 362.
- Jeong, S.W., Choi, Y.J., 2020. Extremophilic microorganisms for the treatment of toxic pollutants in the environment. *Molecules* 25 (21).
- Johnson, J.S., Spakowicz, D.J., Hong, B.Y., Petersen, L.M., Demkowicz, P., Chen, L., Leopold, S.R., Hanson, B.M., Agresta, H.O., Gerstein, M., Sodergren, E., Weinstock, G.M., 2019. Evaluation of 16S rRNA gene sequencing for species and strain-level microbiome analysis. *Nat. Commun.* 10 (1), 5029.
- Joshi, S., Gokhale, S., 2006. Status of mastitis as an emerging disease in improved and periurban dairy farms in India. *Ann. N. Y. Acad. Sci.* 1081, 74–83.
- Khawbung, J.L., Nath, D., Chakraborty, S., 2020. Drug resistant tuberculosis: a review. *Comp. Immunol. Microbiol. Infect. Dis.* 74, 101574.

- Kumar, N., Sharma, G., Leahy, E., Shome, B.R., Bandyopadhyay, S., Deka, R.P., Shome, R., Dey, T.K., Lindahl, J.F., 2021. Understanding antibiotic usage on small-scale dairy farms in the indian states of Assam and Haryana using a mixed-methods approach-outcomes and challenges. *Antibiotics* (Basel) 10 (9).
- Lewnard, J.A., Lo, N.C., Arinaminpathy, N., Frost, L., Laxminarayan, R., 2020. Childhood vaccines and antibiotic use in low- and middle-income countries. *Nature* 581 (7806), 94–99.
- Liu, Y.Y., Wang, Y., Walsh, T.R., Yi, L.X., Zhang, R., Spencer, J., Doi, Y., Tian, G., Dong, B., Huang, X., Yu, L.F., Gu, D., Ren, H., Chen, X., Lv, L., He, D., Zhou, H., Liang, Z., Liu, J.H., Shen, J., 2015. Emergence of plasmid-mediated colistin resistance mechanism MCR-1 in animals and human beings in China: a microbiological and molecular biological study. *Lancet Infect. Dis.* 16 (2), 161–168.
- Mader, R., Bourelly, C., Amat, J.P., Broens, E.M., Busani, L., Callens, B., Crespo-Robledo, P., Damborg, P., Filippitzi, M.E., Fitzgerald, W., Gronthal, T., Haenni, M., Heuvelink, A., van Hout, J., Kaspar, H., Munoz Madero, C., Norstrom, M., Pedersen, K., Pokludova, L., Dal Pozzo, F., Slowey, R., Urdahl, A.M., Vatopoulos, A., Zafeiroidis, C., Madec, J.Y., 2022. Defining the scope of the european antimicrobial resistance surveillance network in veterinary medicine (EARS-Vet): a bottom-up and one health approach. *J. Antimicrob. Chemother.* 77 (3), 816–826.
- Mahon, B.E., Mintz, E.D., Greene, K.D., Wells, J.G., Tauxe, R.V., 1996. Reported cholera in the United States, 1992–1994: a reflection of global changes in cholera epidemiology. *JAMA* 276 (4), 307–312.
- Manyi-Loh, C., Mamphweli, S., Meyer, E., Okoh, A., 2018. Antibiotic use in agriculture and its consequential resistance in environmental sources: potential public health implications. *Molecules* 23 (4).
- Marshall, B.M., Levy, S.B., 2011. Food animals and antimicrobials: impacts on human health. *Clin. Microbiol. Rev.* 24 (4), 718–733.
- McManus, P.S., 2014. Does a drop in the bucket make a splash? Assessing the impact of antibiotic use on plants. *Curr. Opin. Microbiol.* 19, 76–82.
- Mensah, S.E., Koudande, O.D., Sanders, P., Laurentie, M., Mensah, G.A., Abiola, F.A., 2015. Antimicrobial residues in foods of animal origin in Africa: public health risks. *Rev Sci Tech* 33 (3), 975–986, 987–96.
- Mezoughem, C., 2022. Livestock and Poultry: World Markets and Trade World Production, Markets, and Trade Report.
- Michael, C.A., Dominey-Howes, D., Labbate, M., 2014. The antimicrobial resistance crisis: causes, consequences, and management. *Front. Public Health* 2, 145.
- Moore, P.R., Evenson, A., et al., 1946. Use of sulfasuxidine, streptothricin, and streptomycin in nutritional studies with the chick. *J. Biol. Chem.* 165 (2), 437–441.
- Morrison, D.J., Preston, T., 2016. Formation of short chain fatty acids by the gut microbiota and their impact on human metabolism. *Gut Microbes* 7 (3), 189–200.
- Mozaffarian, D., Rimm, E.B., 2006. Fish intake, contaminants, and human health: evaluating the risks and the benefits. *JAMA* 296 (15), 1885–1899.
- Munk, P., Knudsen, B.E., Lukjancenko, O., Duarte, A.S.R., Van Gompel, L., Luiken, R.E. C., Smit, L.A.M., Schmitt, H., Garcia, A.D., Hansen, R.B., Petersen, T.N., Bossers, A., Ruppe, E., Group, E., Lund, O., Hald, T., Pamp, S.J., Vigre, H., Heederik, D., Wagenaar, J.A., Mevius, D., Aarestrup, F.M., 2018. Abundance and diversity of the faecal resistome in slaughter pigs and broilers in nine European countries. *Nat. Microbiol.* 3 (8), 898–908.
- Nepal, G., Bhatta, S., 2018. Self-medication with antibiotics in WHO Southeast Asian region: a systematic review. *Cureus* 10 (4), e2428.
- Nhung, N.T., Chansiripornchai, N., Carrique-Mas, J.J., 2017. Antimicrobial resistance in bacterial poultry pathogens: a review. *Front. Vet. Sci.* 4, 126.
- Nhung, N.T., Van, N.T.B., Cuong, N.V., Duong, T.T.Q., Nhat, T.T., Hang, T.T.T., Nhi, N.T. H., Kiet, B.T., Hien, V.B., Ngoc, P.T., Campbell, J., Thwaites, G., Carrique-Mas, J., 2018. Antimicrobial residues and resistance against critically important antimicrobials in non-typhoidal salmonella from meat sold at wet markets and supermarkets in Vietnam. *Int. J. Food Microbiol.* 266, 301–309.
- O'Neill, J., 2016. Tackling drug-resistant infections globally: final report and recommendations. In: *Review on Antimicrobial Resistance*.
- Palmer, E.L., Teviotdale, B.L., Jones, A.L., 1997. A relative of the broad-host-range plasmid RSF1010 detected in *erwinia amylovora*. *Appl. Environ. Microbiol.* 63 (11), 4604–4607.
- Reeder, P.T.A.R., 2020. Antibiotic use on crops in low and middle-income countries based on recommendations made by agricultural advisors. *BMC CAB Agric Biosci.* 1, 1.
- Reverter, M., Sarter, S., Caruso, D., Avarre, J.C., Combe, M., Pepey, E., Pouyaud, L., Vega-Heredia, S., de Verdal, H., Gozlan, R.E., 2020. Aquaculture at the crossroads of global warming and antimicrobial resistance. *Nat. Commun.* 11, 1870.
- Sack, D.A., Sack, R.B., Nair, G.B., Siddique, A.K., 2004. Cholera. *Lancet* 363 (9404), 223–233.
- Sánchez, C.R., 2008. Oxytetracycline and gentamicin: two clinically relevant antimicrobials widely used by Costa Rican farmers. Implications of their use outside clinical settings and request for action. *Revista Médica de la Universidad de Costa Rica* 2 (2), 8.
- Satish Kumar, J.B., Yadav, Ajay Kumar, Pegu, S.R., 2020. Pattern of antibiotics use in pigs of North Eastern India. *Indian Journal of Animal Health* 59 (01), 50–54.
- Sawant, A.A., Sordillo, L.M., Jayarao, B.M., 2005. A survey on antibiotic usage in dairy herds in Pennsylvania. *J. Dairy Sci.* 88 (8), 2991–2999.
- Schar, D., Zhao, C., Wang, Y., Larsson, D.G.J., Gilbert, M., Van Boeckel, T.P., 2021. Twenty-year trends in antimicrobial resistance from aquaculture and fisheries in Asia. *Nat. Commun.* 12 (1), 5384.
- Shah, M.S., Qureshi, S., Kashoo, Z., Farooq, S., Wani, S.A., Hussain, M.I., Banday, M.S., Khan, A.A., Gull, B., Habib, A., Khan, S.M., Dar, B.A., 2019. Methicillin resistance genes and in vitro biofilm formation among *Staphylococcus aureus* isolates from bovine mastitis in India. *Comp. Immunol. Microbiol. Infect. Dis.* 64, 117–124.
- Shahbandeh, M., 2022. Number of Pigs Worldwide From 2012 to 2022 (in Million Head). *statista*. *
- Sharma, C., Rokana, N., Chandra, M., Singh, B.P., Gulhane, R.D., Gill, J.P.S., Ray, P., Puniya, A.K., Panwar, H., 2018. Antimicrobial resistance: its surveillance, impact, and alternative management strategies in dairy animals. *Front. Vet. Sci.* 4, 237.
- Sidhu, K.S., 2003. Health benefits and potential risks related to consumption of fish or fish oil. *Regul. Toxicol. Pharmacol.* 38 (3), 336–344.
- Soltan, S.A.S.A.S., MEM, 2012. Antidiabetic and Hypocholesterolemic Effect of Different Types of Vinegar in Rats. *Life Science Journal* 9, 2141–2151.
- Stock, I., Wiedemann, B., 2001. Natural antibiotic susceptibilities of *Edwardsiella tarda*, *E. ictaluri* and *E. hoshinae*. *Antimicrob. Agents Chemother.* 45, 2245–2255.
- Sundin, G.W., Wang, N., 2018. Antibiotic resistance in plant-pathogenic bacteria. *Annu. Rev. Phytopathol.* 56, 161–180.
- Svenson, E.L., Tun, R., Conway, J.H., Legenza, L., Svenson, J.M., Duffy, S., Svenson, J.E., 2021. A one health approach to compare self-prescribed antibiotic use across rural and semi-urban populations in San Lucas Toliman, Guatemala. *J. Health Care Poor Underserved* 32 (4), 1798–1817.
- Tao Zhang, S.T., Peng, Haomin, Luo, Huifeng, Wan, Xuanwu, Wan, Min, Wang, Buyun, 2017. Is business linkage affecting agricultural advisory services? *Intern. J. Agric. Ext. (01)* (5), 59–77.
- Thanner, S., Drissner, D., Walsh, F., 2016. Antimicrobial resistance in agriculture. *MBio* 7 (2), e02227-15.
- Thompson, T., 2022. The staggering death toll of drug-resistant bacteria. *Nature* 31. <https://doi.org/10.1038/d41586-022-00228-x>.
- Torris, C., Smastuen, M.C., Molin, M., 2018. Nutrients in fish and possible associations with cardiovascular disease risk factors in metabolic syndrome. *Nutrients* 10 (7).
- Van Boeckel, T.P., Brower, C., Gilbert, M., Grenfell, B.T., Levin, S.A., Robinson, T.P., Teillant, A., Laxminarayan, R., 2015. Global trends in antimicrobial use in food animals. *Proc. Natl. Acad. Sci. U. S. A.* 112 (18), 5649–5654.
- Van Boeckel, T.P., Pires, J., Silvestre, R., Zhao, C., Song, J., Criscuolo, N.G., Gilbert, M., Bonhoeffer, S., Laxminarayan, R., 2019. Global trends in antimicrobial resistance in animals in low- and middle-income countries. *Science* 365 (6459).
- Walia, S., Rana, S.W., Maue, D., Rana, J., Kumar, A., Walia, S.K., 2012. Prevalence of multiple antibiotic-resistant gram-negative bacteria on bagged, ready-to-eat baby spinach. *Int. J. Environ. Health Res.* 23 (2), 108–118.
- Walsh, T.R., 2018. A one-health approach to antimicrobial resistance. *Nat. Microbiol.* 3, 854–855.
- Wang, Y., Zhang, R., Li, J., Wu, Z., Yin, W., Schwarz, S., Tyrrell, J.M., Zheng, Y., Wang, S., Shen, Z., Liu, Z., Liu, J., Lei, L., Li, M., Zhang, Q., Wu, C., Wu, Y., Walsh, T. R., Shen, J., 2017. Comprehensive resistome analysis reveals the prevalence of NDM and MCR-1 in Chinese poultry production. *Nat. Microbiol.* 2, 16260.
- Wegener, H.C., Aarestrup, F.M., Jensen, L.B., Hammerum, A.M., Bager, F., 1999. Use of antimicrobial growth promoters in food animals and enterococcus faecium resistance to therapeutic antimicrobial drugs in Europe. *Emerg. Infect. Dis.* 5 (3), 329–335.
- WHO, 2011. Report on the Burden of Endemic Health Care-associated Infection Worldwide. World Health Organization, Geneva.
- WHO, 2015. Global Action Plan on Antimicrobial Resistance. World Health Organization, Geneva.
- WHO, 2019. Antimicrobial resistance. In: *Seventy-second World Health Assembly WHA72.5*.
- Wunsch, N.-G., 2022. Share of Vegetarian Worldwide in 2021, Select Country. *Statistica.com*.
- Yamaguchi, M., Win, H.P.M., Higashi, K., Ono, M., Hirose, Y., Motooka, D., Okuzaki, D., Aye, M.M., Htun, M.M., Thu, H.M., Kawabata, S., 2021. Epidemiological analysis of pneumococcal strains isolated at Yangon Children's Hospital in Myanmar via whole-genome sequencing-based methods. *Microb. Genom.* 7 (2).
- Yau, S., Liu, X., Djordjevic, S.P., Hall, R.M., 2010. RSF1010-like plasmids in Australian salmonella enterica serovar typhimurium and origin of their sul2-strA-strB antibiotic resistance gene cluster. *Microb. Drug Resist.* 16 (4), 249–252.